A river-network exposure model with a Whittle–Matérn breeding field for onchocerciasis prevalence in Sudan and South Sudan

1 Problem statement

Onchocerciasis is transmitted by blackflies of the genus *Simulium*, whose larvae develop in fast-flowing water. The exposure of a village is therefore generated at breeding sites on the river network, possibly several kilometres away, and reaches the village through the dispersal of flies, while the efficiency with which exposure is converted into infection depends on climate. In earlier work we expressed the force of infection as the product of an exposure computed from the river network and an intensity of transmission depending on aridity. With the suitability of a river described by its discharge alone, the fit required a quadratic effect of discharge and a change point in the effect of aridity, and both features were traced to a single focus beside the Nile rapids near Abu Hamad that discharge could not distinguish from slow stretches of the same river. Here we instead let the suitability of the rivers vary along the network through a Gaussian Whittle–Matérn field, add a village effect for residual variation, and simplify the fixed effects to a linear effect of log discharge on breeding and a linear effect of log aridity on the intensity of transmission.

2 Data

The outcome is the number of individuals with palpable onchocercal nodules among those examined in 900 villages of Sudan and South Sudan surveyed by the Rapid Epidemiological Mapping of Onchocerciasis (REMO). The river network is HydroRIVERS (Lehner and Grill, 2013), restricted to reaches with a long-term average discharge of at least 0.1 m^3/s; it contains 47,588 reaches and 249,403 km of river, and its topology, the reach into which each reach flows, is used to join the reaches at confluences. Aridity is the ratio of annual precipitation to reference evapotranspiration from version 3.1 of the Global Aridity Index (Zomer et al., 2022), averaged within 2 km of each village.

3 Model

Let Γ denote the river network and $x_1, \ldots, x_m$ the villages, with n_i individuals examined and y_i found positive at x_i. The breeding intensity at a point $s \in \Gamma$ is

$$w(s) = \exp\{\theta \log Q(s) + u(s)\},$$

where $Q(s)$ is the discharge of the reach and u is a zero-mean Gaussian field on Γ describing the suitability of the river for breeding beyond what discharge explains. The exposure of a location x is the breeding within reach, weighted by straight-line distance,

$$E(x) = \int_\Gamma \frac{1}{c\,\phi}\, k\!\left(\frac{\|x - s\|}{\phi}\right) w(s)\,\mathrm{d}s, \qquad k(t) = (1-t)_+^4(1+4t),$$

with the Wendland function k (Wendland, 1995), $c = 2/3$ and the dispersal range fixed at $\phi = 40$ km from entomological evidence (Carnaghi et al., 2024). The intensity of transmission is

$$T(x) = \exp\{\beta_0 + \beta_1 \log \mathrm{AI}(x) + Z(x)\},$$

with independent village effects $Z(x_i) \sim N(0, \sigma_Z^2)$, and

$$Y_i \sim \mathrm{Bin}\big(n_i, 1 - \exp\{-E(x_i)\,T(x_i)\}\big).$$

The field u is the Whittle–Matérn field on the metric graph Γ with smoothness $\alpha = 1$ (Bolin et al., 2024), the solution of $(\kappa^2 - \Delta_\Gamma)^{1/2}\tau u = \mathcal{W}$ with Kirchhoff conditions at the vertices. Along a reach far from any vertex it is a stationary Ornstein–Uhlenbeck process with variance σ_u^2 and correlation $\exp(-\kappa d)$ at distance d along the river, and we report its range as $\rho_u = 2/\kappa$, the distance at which this correlation is $e^{-2} \approx 0.14$. Because the field is Markov, its values at the vertices of the graph have a sparse precision matrix with an explicit form: an edge of length ℓ contributes

$$\frac{1}{\sigma_u^2(1 - e^{-2\kappa\ell})} \begin{pmatrix} (1 + e^{-2\kappa\ell})/2 & -e^{-\kappa\ell} \\ -e^{-\kappa\ell} & (1 + e^{-2\kappa\ell})/2 \end{pmatrix},$$

and subdividing an edge by additional vertices leaves the field unchanged. The Kirchhoff conditions make the variance twice σ_u^2 at the upstream ends of the network and smaller than σ_u^2 at confluences, so that information is shared across the reaches that meet at a confluence.

On the log scale, $\log \lambda(x) = \log E(x) + \log T(x)$. The two random terms enter different components and have different spatial structure: u varies along the rivers and reaches a village only after smoothing by the dispersal kernel, so it is shared by the villages that draw on the same stretches of river, whereas Z is specific to each village.

4 Estimation

The network is represented by a graph with a vertex at each end of every reach and additional vertices so that no edge is longer than 5 km, giving 75,119 vertices and 74,810 edges. The exposure integral is computed by quadrature over 148,571 nodes at most 2 km apart, and u at each node is the linear interpolation of its values at the two ends of the edge. The latent vector, consisting of β_0, β_1, θ, the values of u at the vertices and the village effects, is integrated out by a Laplace approximation, with diffuse Gaussian priors (standard deviation 10) on the regression coefficients centred on their estimates without u and Z. The mode is found by a truncated Newton method (Steihaug, 1983): the exact Hessian of the log posterior, which includes the curvature of $\log E$ in θ and u, is inverted by conjugate gradients preconditioned

by the Gauss–Newton Hessian. The latter is the prior precision plus a term whose rank is the number of villages, and is inverted by the Woodbury identity with one sparse Cholesky factor of the precision of u for each value of the range; one evaluation of the approximate likelihood takes 10–20 seconds. The standard deviations σ_u and σ_Z are estimated by maximising the approximate marginal likelihood, and the range ρ_u by profiling it over a grid. Standard errors of the regression coefficients are from the Gauss–Newton Hessian at the mode, conditional on the variance parameters. Computations use the R package `odispers`.

5 Results

Table 1 compares the models. Without latent effects, the linear model fits far worse than the earlier spline and quadratic specification (maximised log-likelihoods -2851.53 and -2609.83), which is why those features were needed. Adding the village effect raises the approximate log-likelihood to -1613.40, with a large standard deviation, $\sigma_Z = 2.21$, so most of the variation between villages is not explained by the fixed effects. Adding the breeding field raises it further, by 248 units at the best range, and halves the village effect to $\sigma_Z = 0.89$: much of the apparent heterogeneity between villages is variation along the rivers, shared by villages that draw on the same stretches. The profile over the range of u (Figure 1) is maximal at about 800 km, with $\sigma_u = 2.76$. A range of several hundred kilometres along the network means that u describes differences between rivers and sub-basins rather than between neighbouring reaches.

Table 1: Approximate (Laplace) log-likelihoods, with the regression coefficients integrated out under diffuse priors, and estimated standard deviations of the breeding field and of the village effect. Dispersal range $\phi = 40$ km in all models.

Model	Log-likelihood	σ_u	σ_Z
Village effect only	-1613.40	–	2.21
u, range 100 km, + village effect	-1400.91	2.80	0.80
u, range 200 km, + village effect	-1386.10	2.65	0.81
u, range 400 km, + village effect	-1372.40	2.55	0.86
u, range 800 km, + village effect	-1365.36	2.76	0.89
u, range 1600 km, + village effect	-1368.70	3.18	0.92

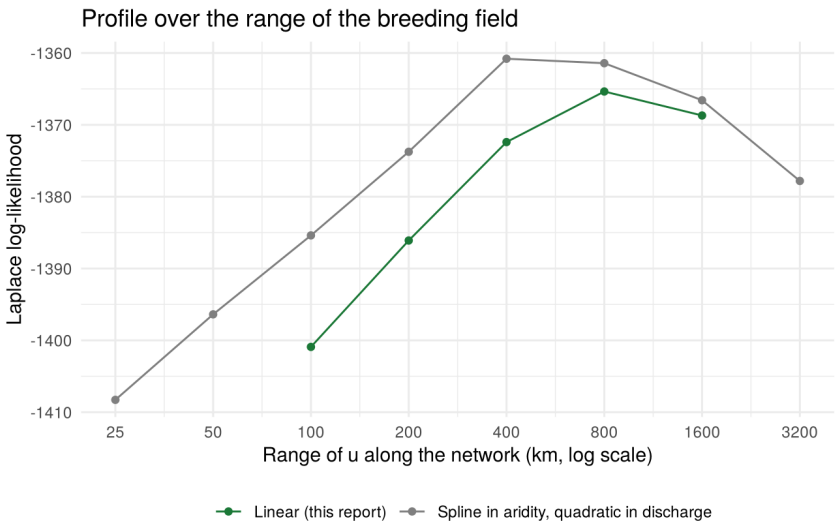


Figure 1: Approximate log-likelihood as a function of the range of the breeding field, for the linear model of this report and for the model with a spline in log aridity and a quadratic in log discharge.

Table 2 reports the regression coefficients. With the breeding field in the model, the effect of aridity is monotone and positive, $\beta_1 = 1.41$ (0.28): doubling the aridity index multiplies the force of infection by $2^{1.41} \approx 2.7$. The effect of discharge is negative, $\theta = -0.29$ (0.11), so that a tenfold increase in discharge roughly halves the breeding intensity per kilometre of river, consistent with small, fast-flowing streams being the main breeding habitat. The standard errors are two to five times larger than without latent effects, because they now account for the residual variation along the rivers and between villages.

Table 2: Estimates (standard errors) of the linear model. Maximum likelihood without latent effects; posterior mode and Gauss–Newton standard errors, conditional on the variance parameters, with the latent effects.

	No latent effects	Village effect only	u (800 km) + village effect
Intercept β_0	−3.33 (0.20)	−3.34 (0.19)	−3.12 (0.37)
log AI, β_1	1.47 (0.05)	1.20 (0.13)	1.41 (0.28)
log Q, θ	−0.94 (0.13)	−0.23 (0.12)	−0.29 (0.11)
σ_u	–	–	2.76
σ_Z	–	2.21	0.89

Figure 2 shows the mode of u on the rivers within reach of a village. The largest values are on the Nile between Abu Hamad and the confluence with the Atbara, the stretch where the river crosses the Fourth and Fifth Cataracts and where the only infected desert villages are found, while the slow stretches of the Nile further north and south and the White Nile have negative values. In South Sudan the field is positive over the dense networks of small streams in the west and centre and in the far south-east. The field therefore identifies the river features that discharge cannot describe, and this is what allows the effects of discharge and aridity to be

linear: in the earlier model without u, the Abu Hamad focus produced the change point in the aridity effect, with a negative slope in the driest conditions, and the increase in breeding for the largest rivers.

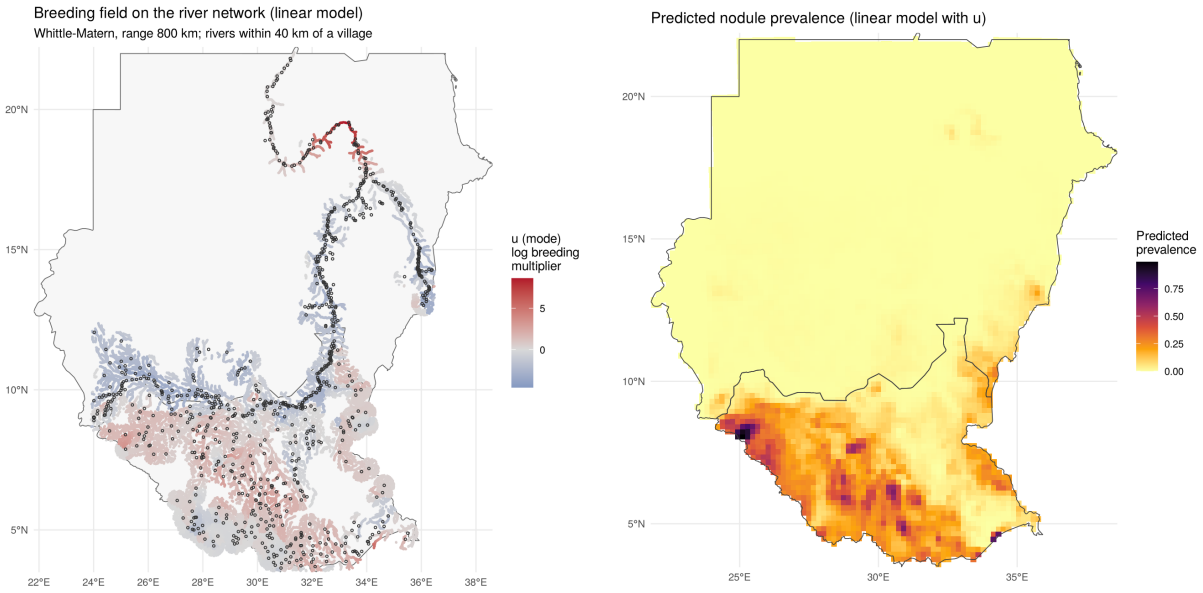


Figure 2: Left: mode of the breeding field u (log multiplier of the breeding intensity) on the rivers within 40 km of a village; circles are the villages. Right: predicted prevalence on a 20 km grid, at the mode of u and with the village effect set to zero.

The predicted prevalence (Figure 2, right) is close to zero across the desert and the Sahel, except along the Nile near Abu Hamad, and is highest in the south-west and centre of South Sudan and in the far south-east. Observed and predicted prevalence at the villages have a correlation of 0.88, against 0.58 for the linear model without latent effects (Figure 3), and the prevalence by aridity class is reproduced closely (Table 3). These are in-sample comparisons in which u has been estimated from the same villages, so they overstate predictive performance; cross-validation is needed for an honest assessment. The predictions underestimate the highest prevalences, because they are evaluated at the mode with the village effect set to zero, which gives the median rather than the mean force of infection.

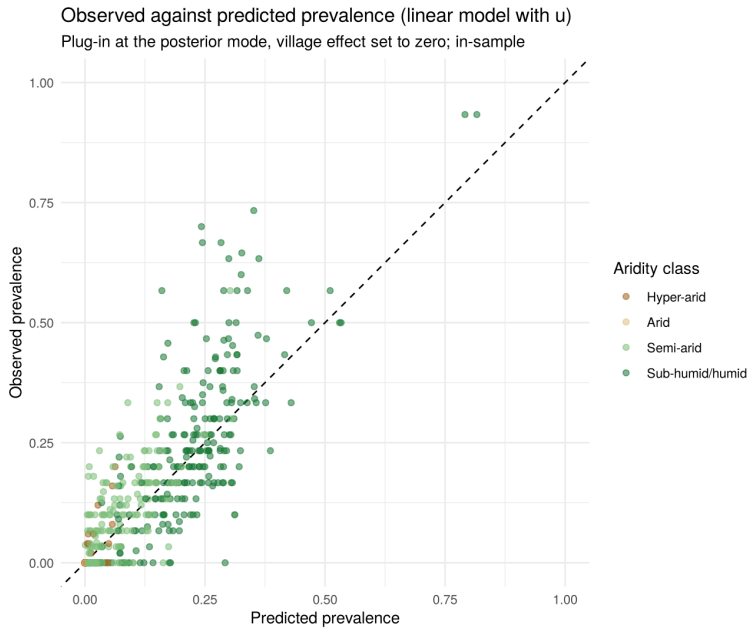


Figure 3: Observed against predicted prevalence at the 900 villages for the linear model with the breeding field (in-sample).

Table 3: Observed and predicted prevalence by aridity class (totals of positives over totals examined).

Aridity class	Observed	Predicted, no latent effects	Predicted, with u
Hyper-arid (AI < 0.05)	0.005	0.001	0.003
Arid (0.05–0.20)	0.000	0.005	0.000
Semi-arid (0.20–0.50)	0.033	0.055	0.028
Sub-humid and humid (AI > 0.50)	0.252	0.197	0.228

If the breeding field accounts for the spatial structure in the data, the village effects Z should be spatially independent once u is in the model, and a further Gaussian process on the plane would not be needed. We assessed this with the empirical variogram of the predicted village effects, the posterior modes of $Z(x_i)$, in distance bins up to 300 km, and with an envelope of spatial independence obtained by randomly permuting the predicted effects among the village locations 999 times; the band between the 2.5th and 97.5th percentiles of the permuted variograms is the range expected in each bin when there is no spatial correlation. A global test combines the bins through the sum of squared standardised differences between the observed variogram and the mean of the permuted variograms, with the p-value given by the rank of the observed statistic among the permuted ones.

Figure 4 shows the result for the model with the village effect only and for the model with u. Without u, the variogram increases steadily from about 0.9 at short distances to 2.0 at 300 km and lies far below the envelope up to about 250 km, so that the village effects are strongly spatially correlated (global test $p = 0.001$, the smallest attainable with 999 permutations); in this model the village effect is absorbing the variation that u represents. With u, the variance

of the predicted village effects falls from 2.17 to 0.18 and the variogram is flat and inside the envelope at every distance (global test $p = 0.155$). At the shortest distances it lies close to the upper limit of the envelope, which is the opposite of positive correlation: neighbouring villages draw on the same stretches of river and share u, so their own effects are estimated as deviations around it. We conclude that the residual variation between villages has no spatial structure, and keep Z as an independent effect. Because the predicted effects are shrunk by different amounts depending on the number examined and on the information about u, they are not exactly exchangeable, so the permutation test is a diagnostic rather than a formal test of the model.

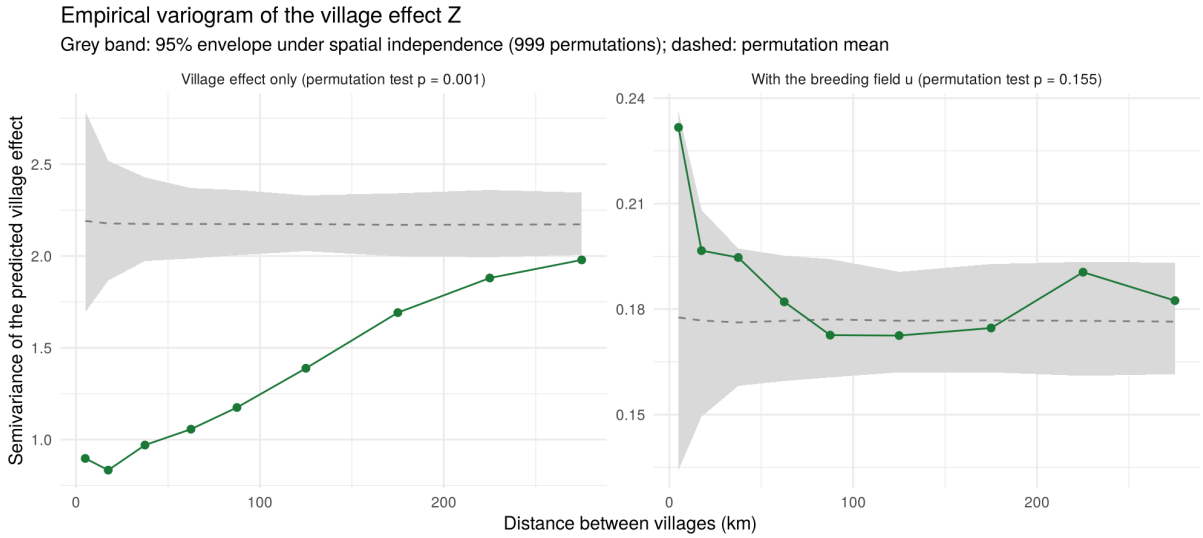


Figure 4: Empirical variogram of the predicted village effect Z for the linear model with the village effect only (left) and with the breeding field u (right). Grey band: 95% envelope under spatial independence from 999 random permutations of the predicted effects among the villages; dashed line: mean of the permuted variograms.

6 Zero inflation

We assessed the predictions of the model of the previous sections by ten-fold cross-validation with villages assigned to folds at random. In each fold the latent effects were refitted to the training villages, with the variance parameters and the range of u fixed at their estimates from all the data, and for each held-out village the predictive distribution of the number of positives was obtained by combining the Gaussian approximation to the posterior of the latent effects, a new village effect and binomial sampling. The assessment was split in two parts, as in a hurdle model: the ability to separate villages without positives from villages with at least one, through the predicted probability $P(Y \geq 1)$, and the calibration of the predicted number of positives among the villages with at least one, through the non-randomised probability integral transform (PIT) of Czado et al. (2009) applied to the predictive distribution conditional on $Y \geq 1$. Under calibration the PIT histogram is flat; its expected variation was obtained by simulating counts from the predictive distributions.

The model separated zero and positive villages well (area under the ROC curve 0.94) but

predicted too few villages without positives, 480 against the 550 observed, almost all of the deficit being in the semi-arid class (219 expected against 275). Among villages with positives, the PIT histogram was hump-shaped, with too few values near 0 and 1 (Figure 6, left), so the predictive distributions were too wide. The two findings have a common explanation: the village effect, with a standard deviation of 0.89, must be wide enough to produce the many villages without positives, which makes it too wide for the villages where transmission occurs. This is the signature of zeros that arise from a separate process.

We therefore let each village lie inside or outside a zone of local transmission. Outside, no individual can be infected; inside, the number of positives is binomial as before. The probability of lying inside a zone is determined by the exposure from the river network,

$$\pi(x) = 1 - \exp\{-\alpha E(x)\},$$

which is the probability that at least one breeding site able to sustain transmission is within reach if such sites occur along the rivers in proportion to the breeding intensity. Within a zone, the force of infection is

$$\lambda(x) = E(x)^{\gamma} \exp\{\beta_0 + \beta_1 \log \mathrm{AI}(x) + Z(x)\},$$

so that

$$P(Y_i = 0) = 1 - \pi(x_i) + \pi(x_i)\{1 - p(x_i)\}^{n_i}, \qquad P(Y_i = y) = \pi(x_i) \binom{n_i}{y} p(x_i)^y \{1 - p(x_i)\}^{n_i - y} \; (y > 0),$$

with $p(x) = 1 - \exp\{-\lambda(x)\}$. The exponent γ determines how far exposure, beyond deciding whether a village is in a zone, also scales the intensity of transmission within it: with $\gamma = 0$ the rivers determine only whether transmission occurs and climate alone determines its intensity, with $\gamma = 1$ exposure enters the force of infection as in the model without zero inflation, and as $\alpha \to \infty$ that model is recovered. Exposure, with its field u (range 800 km), and the village effect are as before. The latent effects are integrated out by the same Laplace approximation, extended to the two predictors $\log E(x_i)$ and $\beta_0 + \beta_1 \log \mathrm{AI}(x_i) + Z(x_i)$ of each village; for this likelihood the Hessian used in the approximation is the observed information of the two predictors, with eigenvalues bounded away from zero. The parameters α, γ, σ_u and σ_Z maximise the approximate marginal likelihood.

Table 4 compares the fits. Zero inflation raises the approximate log-likelihood by 103 units, for two additional parameters, and the estimate $\gamma = 0.19$ is clearly between the two extremes, both of which are rejected (by 37 units for $\gamma = 0$ and 57 units for $\gamma = 1$). The rivers therefore mainly determine whether a village lies in a zone of transmission, while within a zone exposure has a modest effect on intensity: doubling the exposure multiplies the force of infection by $2^{0.19} \approx 1.14$. The village effect is halved, from 0.89 to 0.50, now that it no longer has to produce the excess of zeros, while the field u becomes more variable ($\sigma_u = 4.5$), since it now maps the zones of transmission along the network.

Table 4: Models with and without zero inflation (range of u 800 km). Approximate log-likelihoods with the regression coefficients integrated out under diffuse priors; estimates (standard errors) of the regression coefficients at the posterior mode, conditional on the variance parameters.

	No zero inflation	Zero-inflated		
		γ estimated	$\gamma = 0$	$\gamma = 1$
Log-likelihood	−1365.36	−1262.12	−1298.82	−1319.04
α	–	0.28	0.31	0.28
γ	1	0.19	0	1
σ_u	2.76	4.48	3.61	1.86
σ_Z	0.89	0.50	0.66	0.51
Intercept β_0	−3.12 (0.37)	−1.98 (0.07)	−1.22 (0.06)	−3.60 (0.11)
log AI, β_1	1.41 (0.28)	0.25 (0.06)	0.45 (0.06)	0.04 (0.09)
log Q, θ	−0.29 (0.11)	−0.43 (0.16)	−0.33 (0.14)	−0.22 (0.07)

The estimated probability of lying in a zone of transmission (Figure 5) averages 0.08 for the hyper-arid villages, 0.01 for the arid, 0.34 for the semi-arid and 0.95 for the sub-humid and humid villages. It is close to zero along almost the whole of the Nile and the White Nile in Sudan and across the northern Sahel, between 0.7 and 0.9 at the villages near Abu Hamad where infection was found in the desert, intermediate along the White Nile through the Sudd swamps, whose slow water offers little breeding habitat, and close to one over most of South Sudan.

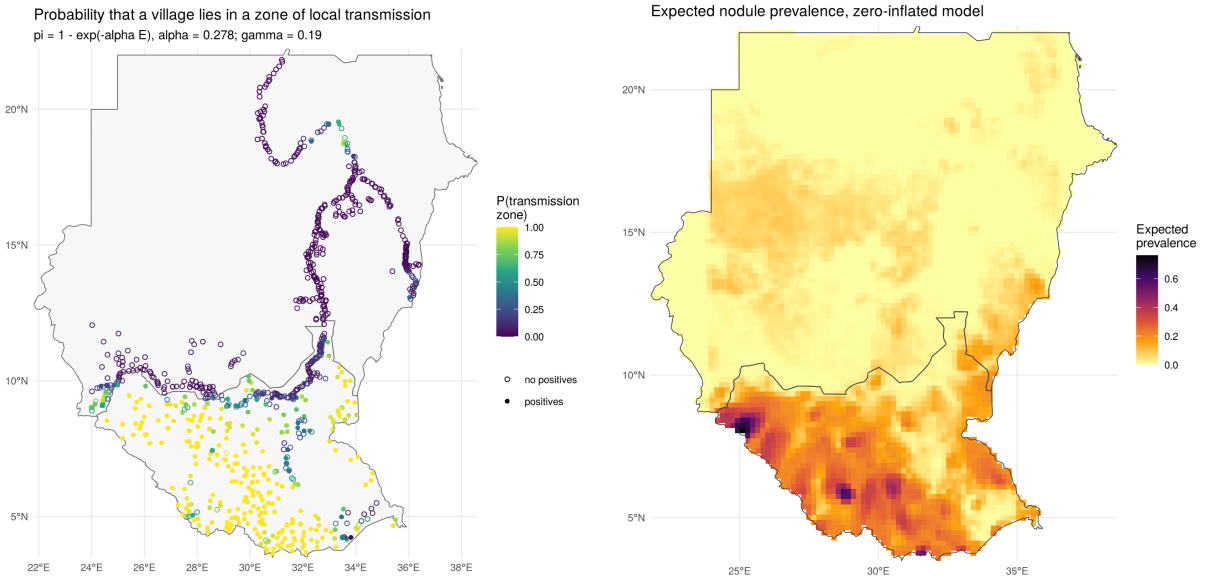


Figure 5: Zero-inflated model with γ estimated. Left: probability that each village lies in a zone of local transmission (filled symbols: villages with positives). Right: expected prevalence $\pi(x)p(x)$ on a 20 km grid, at the posterior mode with the village effect set to zero.

The same random-fold cross-validation shows that zero inflation removes both deficiencies (Table 5 and Figure 6). Among villages with positives, the PIT histogram becomes flat and lies within its expected variation, so the predicted numbers of positives are now calibrated. The predicted number of villages without positives rises from 480 to 532, against 550 observed, and in the semi-arid class from 219 to 265, against 275; the reliability diagram of $P(Y \geq 1)$ lies on the diagonal; and the Brier score and the mean logarithmic score both improve. The area under the ROC curve is practically unchanged, so the gain is in calibration rather than in discrimination. The one remaining discrepancy is a modest excess of predicted zeros among the humid villages, 28 against 14 observed.

Table 5: Ten-fold cross-validation with random folds: discrimination between villages with and without positives, overall predictive score, and observed and expected numbers of villages without positives, in total and by aridity class.

	No zero inflation	Zero-inflated
Area under the ROC curve, $P(Y \geq 1)$	0.939	0.943
Brier score, $P(Y \geq 1)$	0.102	0.089
Mean logarithmic score of the count	−1.452	−1.342
Villages without positives (550 observed)	479.8	532.5
Hyper-arid (154 observed)	144.0	141.1
Arid (107 observed)	98.6	97.7
Semi-arid (275 observed)	219.2	265.3
Sub-humid and humid (14 observed)	18.1	28.5

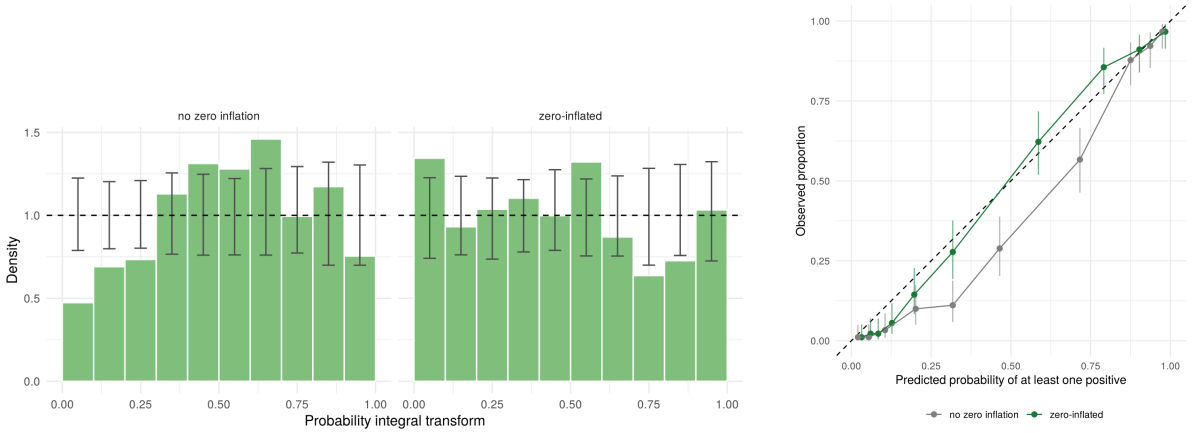


Figure 6: Ten-fold cross-validation with random folds. Left: non-randomised PIT among villages with at least one positive, for the models without and with zero inflation; bars show the 95% range expected under calibration. Right: reliability of the predicted probability of at least one positive (deciles of the predicted probability, with 95% intervals).

One feature of the zero-inflated model requires care in interpretation and prediction. The effect of aridity on the intensity of transmission falls from 1.41 to 0.25: the contrast between the arid north and the humid south is now carried mainly by the probability of a zone of

transmission, and therefore by the field u, which has no information that this contrast follows the climate rather than the rivers. Within the surveyed river systems this does not affect the predictions, as the cross-validation shows. Away from them, where u returns to its mean, the probability of a zone is driven by the density of the river network alone, and the model predicts a high probability of transmission zones, and expected prevalences of 5 to 15%, over parts of Darfur and Kordofan without survey data, some of them arid (Figure 7). Such predictions should not be trusted. Letting the climate enter the probability of a zone, $\pi(x) = 1 - \exp\{-\alpha E(x)\mathrm{AI}(x)^{\delta}\}$, would allow aridity to explain the absence of transmission in the north, leave u with the effects of the rivers, and give more credible predictions where there are no data; this extension is being fitted.

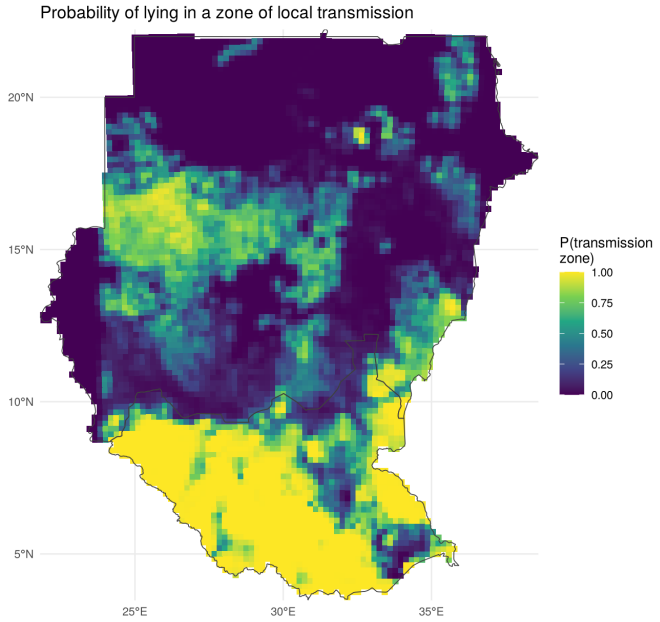


Figure 7: Probability of lying in a zone of local transmission on a 20 km grid (zero-inflated model, γ estimated, at the posterior mode). Away from the surveyed rivers, where u returns to its mean, the probability depends on the density of the river network alone.

7 Comparison with the more flexible specification and discussion

The same model with a spline in log aridity and a quadratic in log discharge reaches an approximate log-likelihood of -1360.80, at a range of 400 km, against -1365.36 for the linear model, a difference of 4.6 units for two additional coefficients and an estimated change point. Since the regression coefficients are integrated out, these marginal likelihoods include a penalty for each coefficient that depends on its prior standard deviation, so the comparison is indicative; but both specifications describe the data similarly once the breeding field is included. In the flexible specification with u, the slope of the aridity effect below the change point is -0.8 with a stan-

dard error of about 0.5 and the quadratic term in discharge is not significant, so its additional features are no longer supported. We therefore prefer the linear model, which has a monotone effect of aridity with a clear interpretation, a single parameter for the effect of river size, and attributes the remaining variation to the rivers and to the villages.

The decomposition of the force of infection into exposure and intensity of transmission is retained and sharpened. The exposure now depends on the rivers through both their discharge and the field u, which acts as a map of the suitability of the river network for breeding, and the intensity of transmission through aridity and a village effect. Places beyond 40 km of any river still have zero exposure. Several points need attention. First, u is informed only on the rivers within reach of a village, so elsewhere its mode is close to zero and its uncertainty large; predictions in unsurveyed basins inherit this uncertainty, which the plug-in maps do not show. Second, the range of u is large and its variance at the ends of the network is twice that in the interior, as implied by the Kirchhoff conditions; a field with smoothness $\alpha = 2$ or a stationary boundary correction could be considered. Third, the field may be partly explained by measured river characteristics, such as channel gradient and the seasonality of flow from RiverATLAS, which would turn part of u into interpretable covariates. Fourth, the absence of spatial correlation in the village effects once u is included indicates that a Gaussian process on the plane in the intensity of transmission is not needed; such a process would in any case be difficult to separate from u, whose contribution to the exposure is smooth over several tens of kilometres after the dispersal kernel is applied. Fifth, the cross-validation shows that the excess of villages without positives is better described by a zero-inflated model, in which the rivers determine whether a village lies in a zone of transmission; its predictions are calibrated within the surveyed river systems, but should include the climate in the probability of a zone before they are used to map areas without data. Finally, uncertainty in predictions should be obtained by sampling from the Laplace approximation rather than shown at the posterior mode.

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